SQL Portfolio Project

Hospital Data Analysis Using PostgreSQL

Project Overview

This project focuses on analyzing hospital data using SQL in PostgreSQL. The objective was to solve real-world business problems by writing SQL queries and extracting meaningful insights from the dataset. The project covers data analysis techniques such as aggregation, filtering, sorting, grouping, and date-based analysis to support data-driven decision-making.

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Tools & Technologies

- PostgreSQL
- SQL
- Microsoft Word

Project Duration

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--Create_Table—

```sql
CREATE TABLE Hospital(
    Hospital_Name VARCHAR(50),
    Location VARCHAR(30),
    Department VARCHAR(20),
    Doctors_Count INT,
    Patients_Count INT,
    Admission_Date DATE,
    Discharge_Date DATE,
    Medical_Expenses DECIMAL(10,2));

SELECT * FROM Hospital;
```

--Import Data Into Hospital--

```sql
COPY
Hospital (Hospital_Name,  Location, Department, Doctors_Count, Patients_Count, Admission_Date, Discharge_Date,  Medical_Expenses))
FROM ''
CSV HEADER;
```

--1. Total Number of Patients--

--Write an SQL query to find the total number of patients across all hospitals--

```sql
SELECT hospital_name, SUM(patients_count) AS total_patients
FROM Hospital
GROUP BY hospital_name;
```

--2. Average Number of Doctors per Hospital--

--Retrieve the average count of doctors available in each hospital--

```
SELECT hospital_name, AVG(doctors_count) AS average_doctors
FROM Hospital
GROUP BY hospital_name;
```

--3. Top 3 Departments with the Highest Number of Patients--

--Find the top 3 hospital departments that have the highest number of patients--

```
SELECT hospital_name, department, patients_count AS highest_patients
FROM Hospital
ORDER BY patients_count DESC
LIMIT 3;
```

--4. Hospital with the Maximum Medical Expenses--

--Identify the hospital that recorded the highest medical expenses--

```
SELECT hospital_name, medical_expenses AS highest_medical_expenses
FROM Hospital
ORDER BY medical_expenses DESC
LIMIT 1;
```

--5. Daily Average Medical Expenses--

--Calculate the average medical expenses per day for each hospital--

```
SELECT hospital_name, AVG(medical_expenses/(discharge_date - admission_date)) AS
daily_avg_expenses
FROM Hospital
GROUP BY hospital_name;
```

--6. Longest Hospital Stay--

--Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date--

```
SELECT patients_count, hospital_name, discharge_date, admission_date,
(discharge_date - admission_date) AS stays_days
FROM Hospital
ORDER BY stays_days DESC;
```

--7. Total Patients Treated Per City--

--Count the total number of patients treated in each city--

```
SELECT location, SUM(patients_count) AS total_patients
FROM Hospital
GROUP BY location;
```

--8. Average Length of Stay Per Department--

--Calculate the average number of days patients spend in each department—

```
SELECT department, AVG(discharge_date - admission_date) AS average_staydays
FROM Hospital
GROUP BY department;
```

--9. Identify the Department with the Lowest Number of Patients--

--Find the department with the least number of patients—

```
SELECT department, SUM(patients_count) AS total_patients
FROM Hospital
GROUP BY department
```

```
ORDER BY total_patients
LIMIT 1;

--10. Monthly Medical Expenses Report--
--Group the data by month and calculate the total medical expenses for each month--
SELECT TO_CHAR(admission_date, 'Month') AS monthly_expense,
SUM(medical_expenses) AS total_expense
FROM Hospital
GROUP BY monthly_expense;

--completed--
```